

作业纸

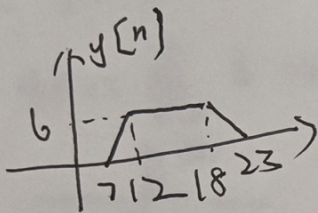
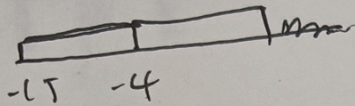
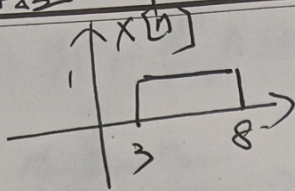
课程名称: 信号与系统

班级: 18级 07449 教学班级: 03152441

姓名: 陈星图

学号: 112024000 第 1 页

2.4:



$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

①: $7 \leq n \leq 12$

$$3 \leq k \leq n-4$$

$$\Rightarrow n-4-3+1 = n-6$$

②: $13 \leq n \leq 18$

$$3 \leq k \leq 8$$

$$8-3+1 = 6$$

③: $19 \leq n \leq 23$

$$n-15 \leq k \leq 8$$

$$8 - (n-15) + 1 = 24 - n$$

$$y[n] = \begin{cases} 0 & n < 7 \\ n-6 & 7 \leq n \leq 12 \\ 6 & 13 \leq n \leq 18 \\ 24-n & 19 \leq n \leq 23 \\ 0 & n > 23 \end{cases}$$

$$n < 7 \Rightarrow n < 3$$

$$7 \leq n \leq 12$$

$$13 \leq n \leq 18$$

$$19 \leq n \leq 23$$

① $t < 3$ $u(t-3) = 0$ $y(t) = 0$

$$t-3 < 0$$

② $3 \leq t < 5$ $u(t-3) = 1$, $t-3 > 0$

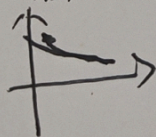
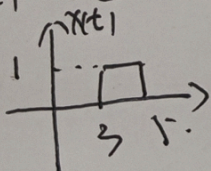
$$y(t) = \int_3^t e^{-3(t-\tau)} d\tau = \frac{1}{3} (1 - e^{-3(t-3)})$$

③ $t \geq 5$ $u(t-3) = 1$

$$y(t) = \int_3^5 e^{-3(t-\tau)} d\tau = \frac{1}{3} (e^{-3(t-3)} - e^{-3(t-5)})$$

2.11: $x(t) = u(t-3) - u(t-5)$

$$h(t) = e^{-3t} u(t)$$



$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_3^5 e^{-3(t-\tau)} u(t-\tau) d\tau$$

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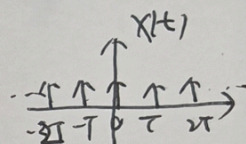
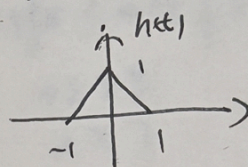
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$$2 \quad y(t) = \begin{cases} 0 & t < 3 \\ \frac{1}{3}(1 - e^{-3(t-3)}) & 3 \leq t < 5 \\ \frac{1}{3}(e^{-3(t-5)} - e^{-3(t-3)}) & t \geq 5 \end{cases}$$

$$1b) : y(t) = \frac{-1+j}{6} e^{-4t} u(t) + \frac{-1-j}{6} e^{-t} e^{j3t} u(t) \\ \therefore y(t) = \frac{1}{6} [e^{-t} (\cos 3t + \sin 3t) - e^{-4t}] u(t)$$

$$1b) : \frac{dx(t)}{dt} = \delta(t-3) - \delta(t-5) \quad 2.23$$

$$g(t) = [\delta(t-3) - \delta(t-5)] h(t) \\ = h(t-3) - h(t-5)$$



$$\therefore g(t) = e^{-3(t-3)} u(t-3) - e^{-3(t-5)} u(t-5)$$

$$\delta(t-a) * h(t) = h(t-a)$$

$$11) \quad 1c) \quad g(t) = \frac{dy(t)}{dt}$$

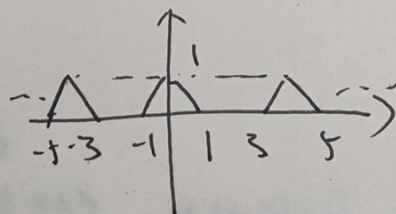
$$1a) : T = 4$$

$$2.17 : \frac{d}{dt} y(t) + 4y(t) = x(t)$$

$$\sum_{k=-\infty}^{+\infty} \delta(t-4k) * h(t) = h(t-4k)$$

$$1a) : sY(s) + 4Y(s) = X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+4}$$

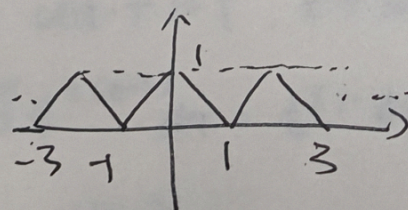


$$X(s) = \frac{1}{s+1-3j}$$

$$Y(s) = \frac{1}{(s+4)(s+1-3j)}$$

$$(b) : T = 2 \quad \sum_{k=-\infty}^{+\infty} h(t-2k)$$

$$y(t) = \frac{1}{6} [t(t+j)e^{-4t} + (t-j)e^{(-1-3j)t}] u(t)$$



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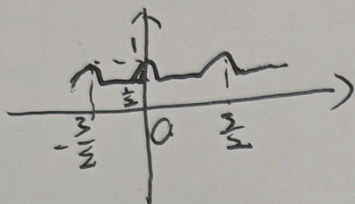
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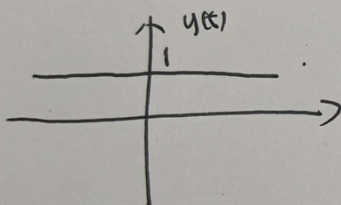
1c1:

$$T = \frac{\pi}{2}, \quad y(t) = \sum_{k=-\infty}^{\infty} h(t - \frac{\pi}{2}k)$$



1d1: $T=1$

$$y(t) = \sum_{k=-\infty}^{\infty} h(t-k)$$



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